Lab 9

DSA

March 27, 2026

Exercise

Implement a Binary Search Tree (BST) in **C/C++**. Your program must support the following operations [CO1, CO2, CO3, CO4]:

- Insertion of a node
- Deletion of a node. It must handle all 3 cases:
  1. The node is a leaf node
  2. The node has a single child
  3. The node has two children
- Inorder traversal

Tasks:

1. Given an array of integers, construct a BST by inserting each element one-by-one into an empty tree
2. Perform the inorder traversal of the BST and print the elements
3. Given a value delete the corresponding node from the BST
4. Print inorder traversal after deletion

Test Case 1

- Insert: 50, 30, 70, 60, 80
- Delete: 30
- Output after deletion: 50 60 70 80

Test Case 2

- Insert: 50, 30, 70, 20
- Delete: 30
- Output after deletion: 20 50 70

Test Case 3

- Insert: 50, 30, 70, 20, 40, 60
- Delete: 50
- Output after deletion: 20 30 40 60 70

Test Case 4

- Insert: 50, 30, 70, 20, 40, 60, 80, 55
- Delete: 50
- Output after deletion: 20 30 40 55 60 70 80

Note: The above test cases correspond to the four deletion scenarios discussed in class using the Transplant operation.

Rubric (10 Marks)

- Correct node insertion logic and BST creation (**2 Marks**)
- Correct inorder traversal implementation (**1 Marks**)
- Correct logic for BST node deletion, covering all 4 test cases (**7 Marks**)